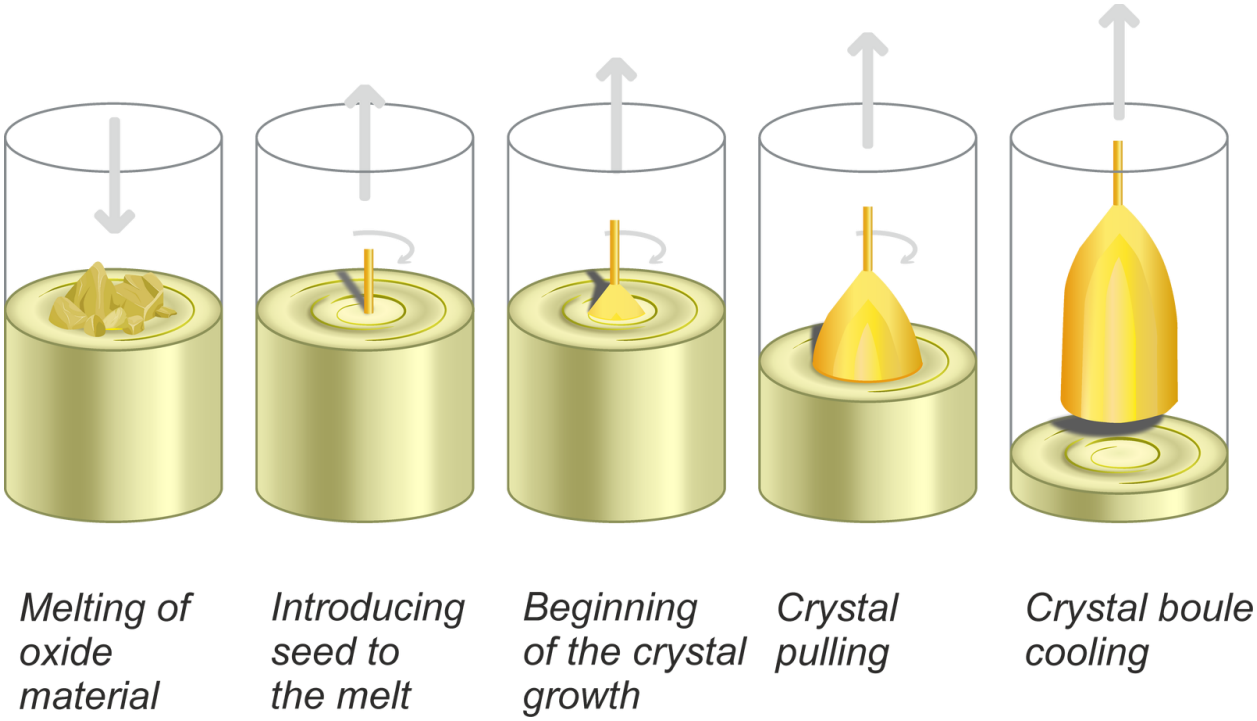
# Czochralski method

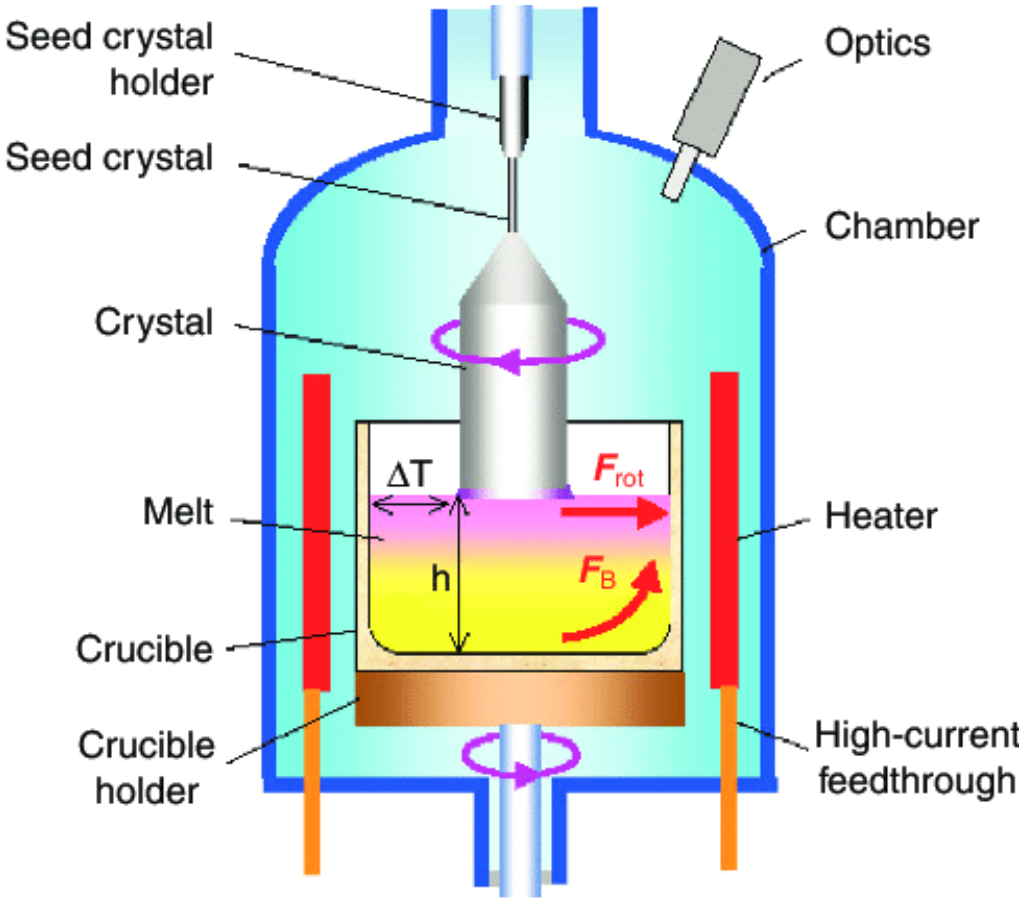
Principle of method

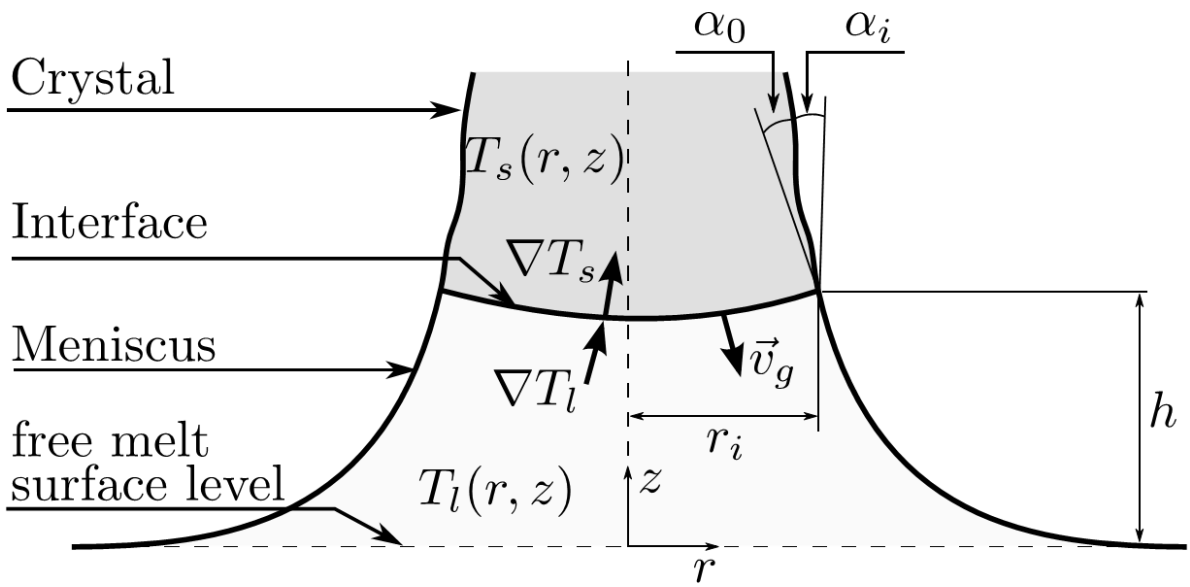
The **Czochralski method** (also referred to as the Czochralski technique or process) is a widely used crystal growth method for producing high-quality single crystals. It is extensively applied to semiconductors such as silicon, germanium, and gallium arsenide, as well as to metals (e.g., palladium, platinum, silver, and gold), salts, and synthetic gemstone materials including garnets and perovskites.

The method is named after Jan Czochralski, who discovered it in 1915 while studying the crystallization of metals. Remarkably, the discovery was accidental: instead of dipping his pen into an inkwell, he immersed it in molten tin and withdrew a thin filament, which was later identified as a single crystal.



Typical construction of the furnace



**How it works (in plain terms)**

1. **Melt the material**
   The raw material (e.g. silicon) is melted in a crucible at high temperature.

2. **Insert a seed crystal**
   A small, well-oriented single crystal (the "seed") is dipped into the melt.

3. **Pull and rotate**
   The seed is slowly pulled upward while rotating.
   At the interface, atoms arrange themselves following the seed's crystal structure.

4. **Grow a single crystal boule**
   A large cylindrical crystal (called a *boule*) forms, maintaining a uniform lattice.

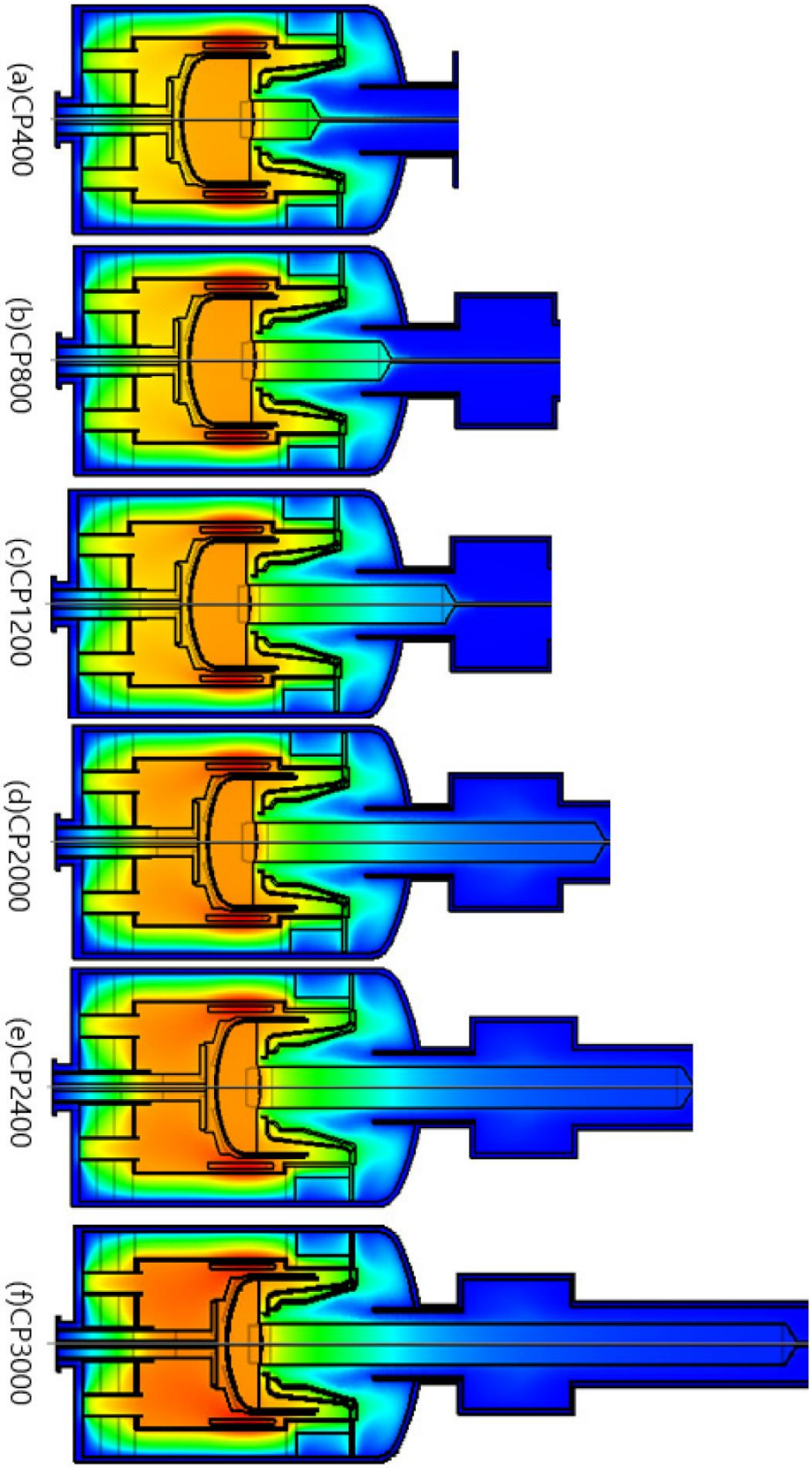
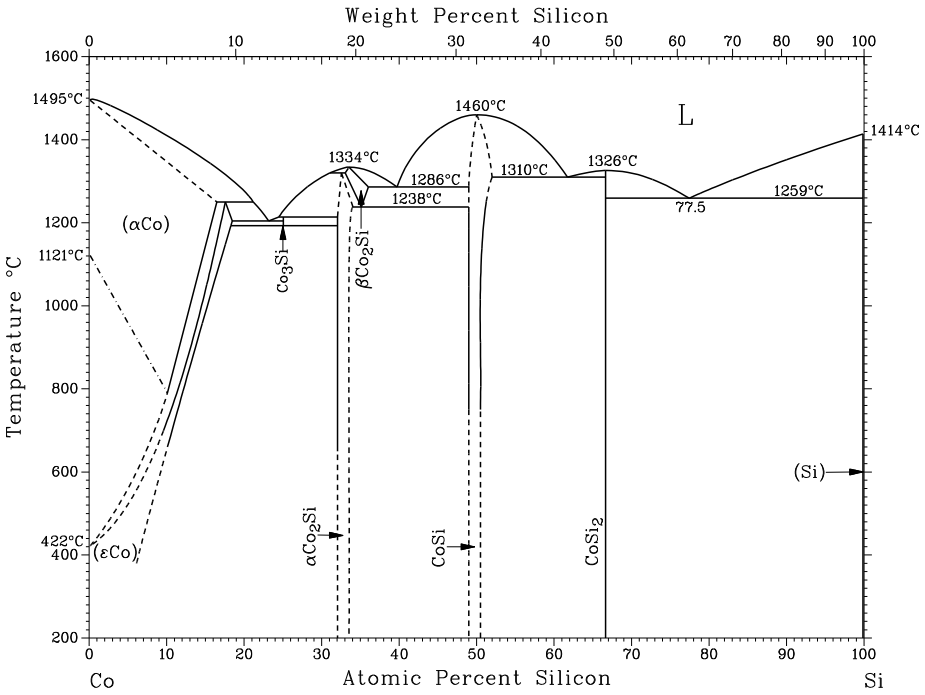
**What creates the concentration gradient?**

At the solid–liquid interface:

- Dopants partition according to the **segregation coefficient**

$$k = \frac{C_s}{C_l}$$

- For most dopants in semiconductors: **k < 1**
  → dopant is **rejected into the melt**


(a)CP400
(b)CP800
(c)CP1200
(d)CP2000
(e)CP2400
(f)CP3000

Assessed Co-Si phase diagram.